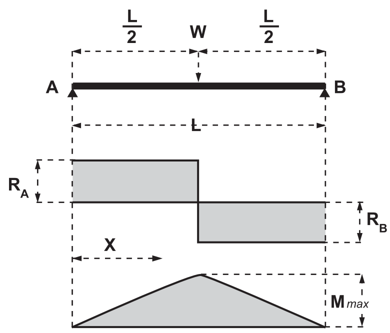
Simply supported beam — Point load at mid-span



Span $= L$

Point load $= W$

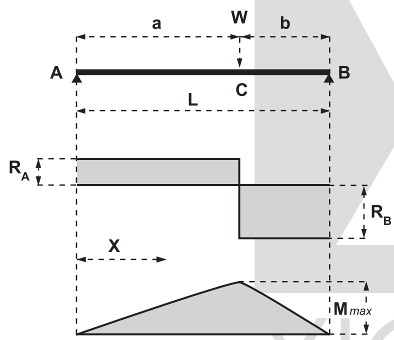
$R_A = R_B = \dfrac{W}{2}$

at mid-span: $M_{max} = \dfrac{WL}{4}$; $\delta_{max} = \dfrac{1}{48} \cdot \dfrac{WL^3}{EI}$

$i_A = i_B = \dfrac{WL^2}{16EI}$

at X from A between A and centre: $M_X = \dfrac{WX}{2}$; $\delta_X = \dfrac{WX}{48EI}(3L^2-4X^2)$; $i_x = \dfrac{W}{16EI}(L^2-4X^2)$

Simply supported beam — Point load at any position



Span $= L$

Point load $= W$

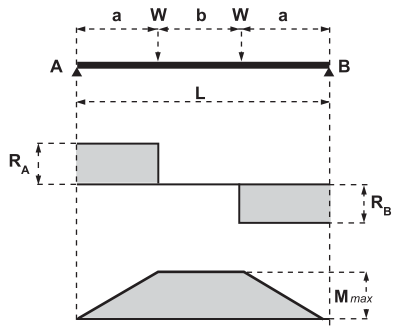
$R_A = \dfrac{Wb}{L}$ $\quad R_B = \dfrac{Wa}{L}$

at C under load: $M_{max} = \dfrac{Wab}{L}$; $\delta_c = \dfrac{Wa^2b^2}{3EIL}$

$i_A = \dfrac{Wab}{6EIL}(L+b)$; $\quad i_B = \dfrac{Wab}{6EIL}(L+a)$

When a>b, δ_{max} is at X from A: $\delta_{max} = \dfrac{Wab(L+b)}{27EIL}\sqrt{3a(L+b)}$; $X = \sqrt{\dfrac{a(L+b)}{3}}$

Simply supported beam — Two equal symmetrical point loads



Span $= L$

Two point loads, each $= W$

$R_A = R_B = W$

M_{max} over length b $= Wa$

δ_{max} at mid-span $= \dfrac{Wa}{24EI}(3L^2-4a^2)$

δ under either load $= \dfrac{Wa^2}{6EI}(3L-4a)$

$i_A = i_B = \dfrac{Wa}{2EL}(L-a)$

If $a=b=\dfrac{L}{3}$, $\delta_{max} = \dfrac{23}{648} \cdot \dfrac{WL^3}{EI}$